

COURSE PROJECT

Title: Smart Pill Bottle Cap

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COURSE PROJECT

SMART PILL BOTTLE CAP

KLE Technological University • ECE Department

SENSE

PROCESS

COMMUNICATE

POWER

ACTUATE

Introduction

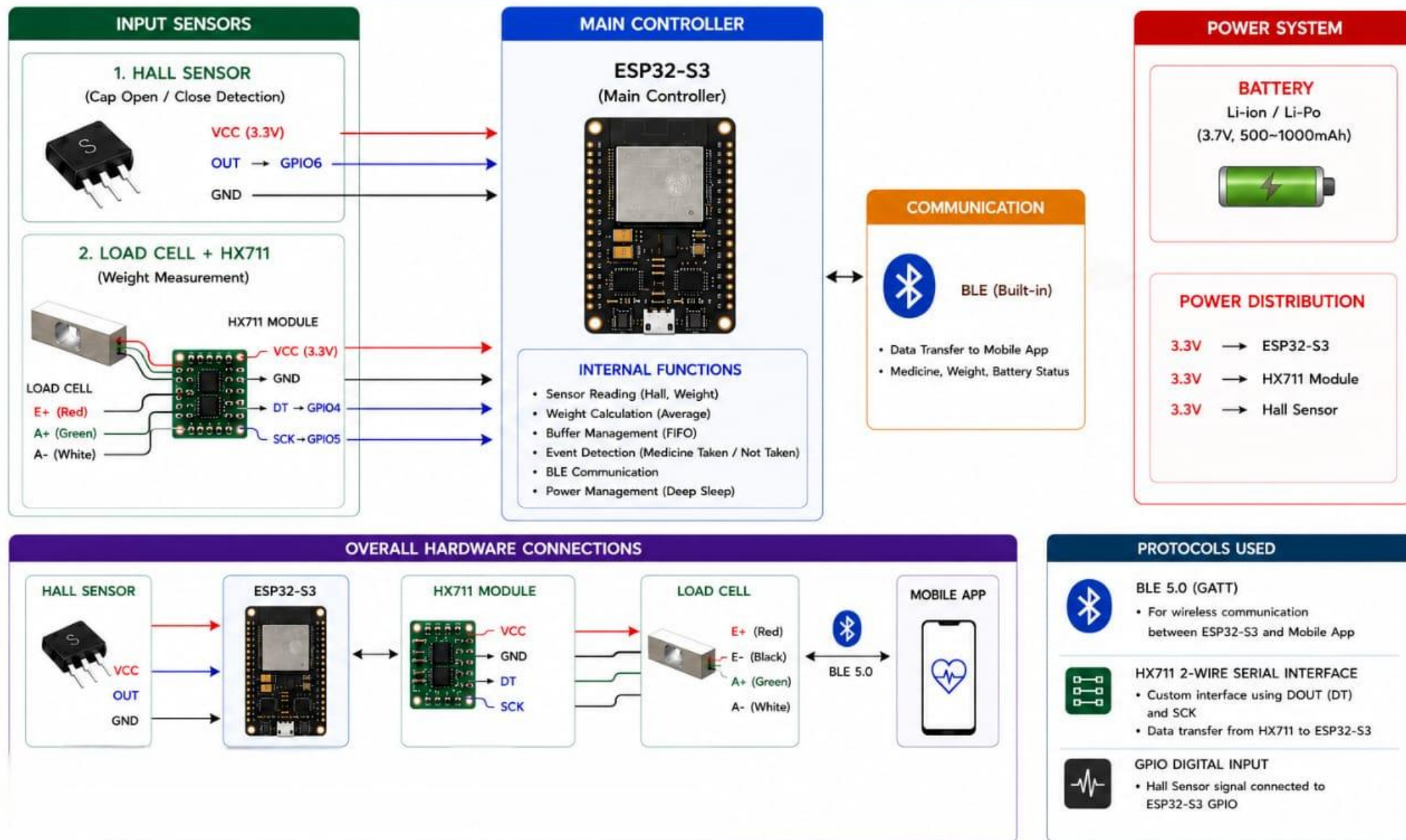
Many people forget to take their medicines on time, especially elderly patients and people with busy schedules. Missing medicines can affect their health and treatment.

The Smart Pill Bottle Cap is a simple IoT-based system that helps users take medicines on time. It uses a Hall sensor to detect bottle opening and a load cell with an HX711 module to measure the bottle weight. An ESP32-S3 processes the data and sends reminders and medicine status to a mobile application using Bluetooth Low Energy (BLE).

The main aim of this project is to improve medicine adherence by providing an easy, reliable, and low-power monitoring system.

SMART PILL BOTTLE CAP – OVERALL HARDWARE ARCHITECTURE

ESP32-S3 BASED SYSTEM (Using BLE Communication)



Hardware Architecture

The Smart Pill Bottle Cap is built around the **ESP32-S3 microcontroller**, which acquires sensor data, processes medication events, and communicates with a mobile application using Bluetooth Low Energy (BLE).

Components

1. ESP32-S3 (Main Controller)

- Central processing unit of the system.
- Reads data from the Hall sensor and HX711 module.
- Calculates tablet count based on weight measurements.
- Detects medication events (tablet taken or missed).
- Transmits notifications to the mobile application via BLE.
- Supports low-power operation for battery-powered applications.

2. Hall Sensor

Function: Bottle cap status detection.

- Detects whether the bottle cap is opened or closed.
- Connected directly to an ESP32 GPIO pin.
- Provides a digital HIGH/LOW signal.

Interface:

- VCC: 3.3 V
- GND
- GPIO Output

3. Load Cell

Function: Weight measurement.

- Measures the total weight of the medicine bottle.
- Converts mechanical force into a small electrical signal.
- Connected to the HX711 amplifier.

4. HX711 Module

Function: Load cell signal conditioning.

- Amplifies the low-level load cell output.
- Converts analog signals into 24-bit digital data.
- Communicates with ESP32 using a two-wire serial interface.

Connections:

- DT (Data)
- SCK (Clock)

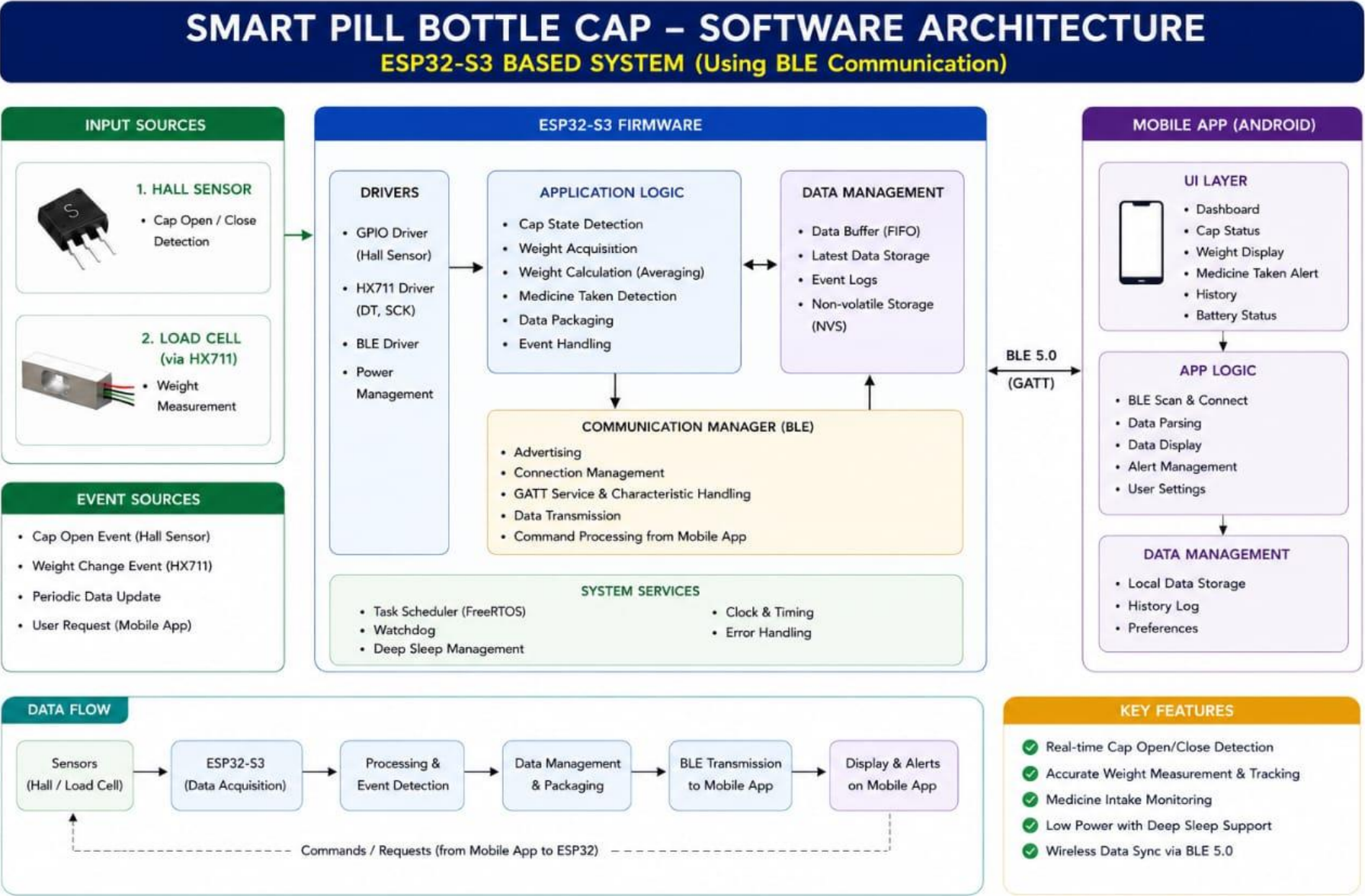
5. BLE Communication

Function: Wireless communication.

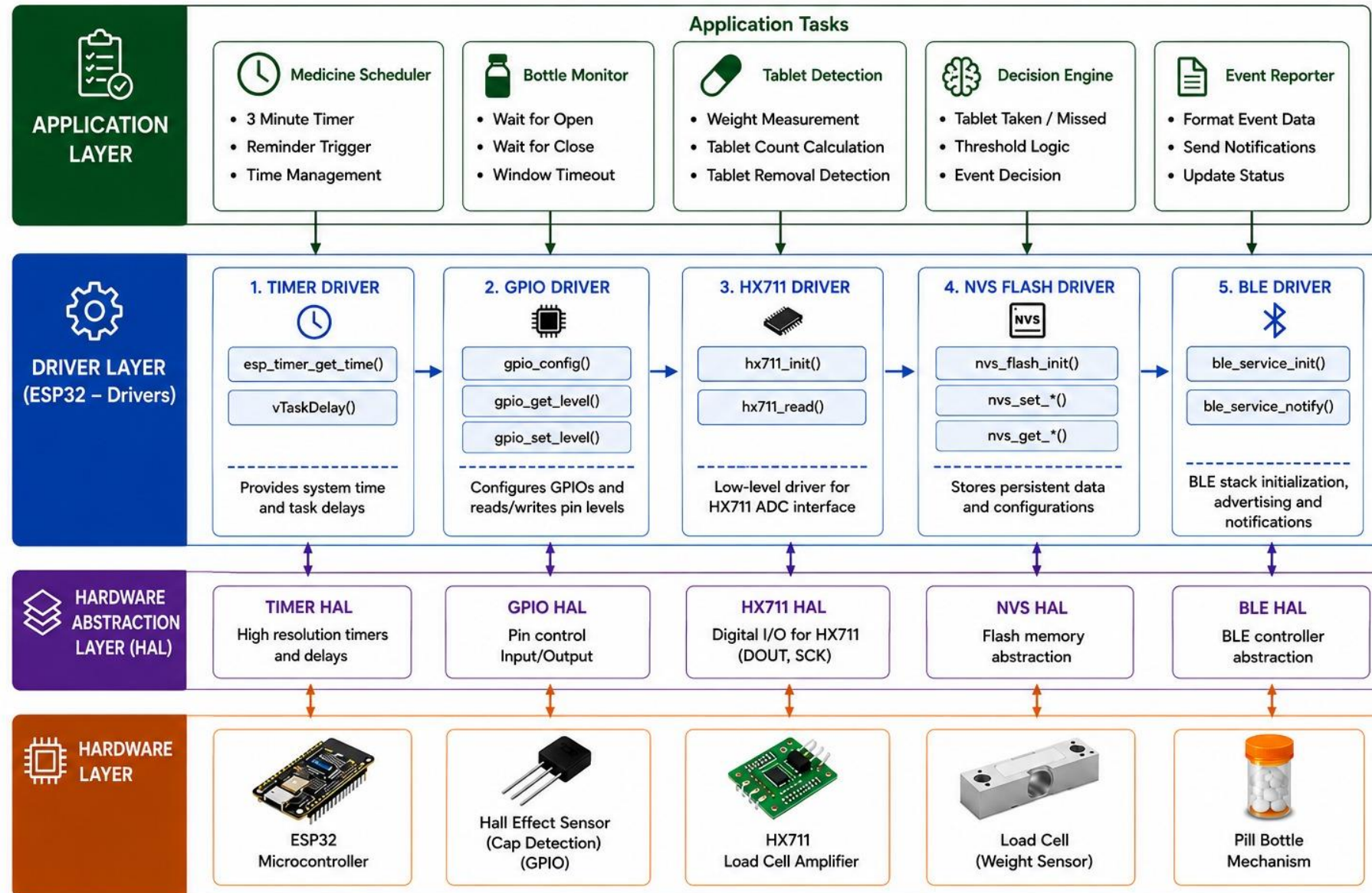
- Built into the ESP32-S3.
- Sends medicine reminders.
- Notifies bottle opening events.
- Reports tablet taken/missed status.
- Transfers weight and tablet count information.

6. Battery Power Supply

- 3.7 V Li-ion/Li-Po battery.
- Supplies power to:
 - ESP32-S3
 - HX711 module
 - Hall sensor



SMART PILL BOTTLE – DRIVER ARCHITECTURE



Driver Architecture

Application Layer

- **Medicine Scheduler** – Manages medication schedules and reminder timings.
- **Bottle Monitor** – Detects bottle cap opening and closing using the Hall Effect Sensor.
- **Tablet Detection** – Measures bottle weight and identifies tablet removal using the Load Cell and HX711.
- **Decision Engine** – Analyzes sensor data to determine whether medication was taken or missed.
- **Event Reporter** – Generates medication status updates and sends notifications via BLE

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Driver Layer

- **Timer Driver** – Handles timing and scheduling operations.
- **GPIO Driver** – Reads Hall Sensor signals and controls I/O pins.
- **HX711 Driver** – Acquires weight measurements from the load cell amplifier.
- **NVS Flash Driver** – Stores schedules, settings, and medication records.
- **BLE Driver** – Manages Bluetooth Low Energy communication.

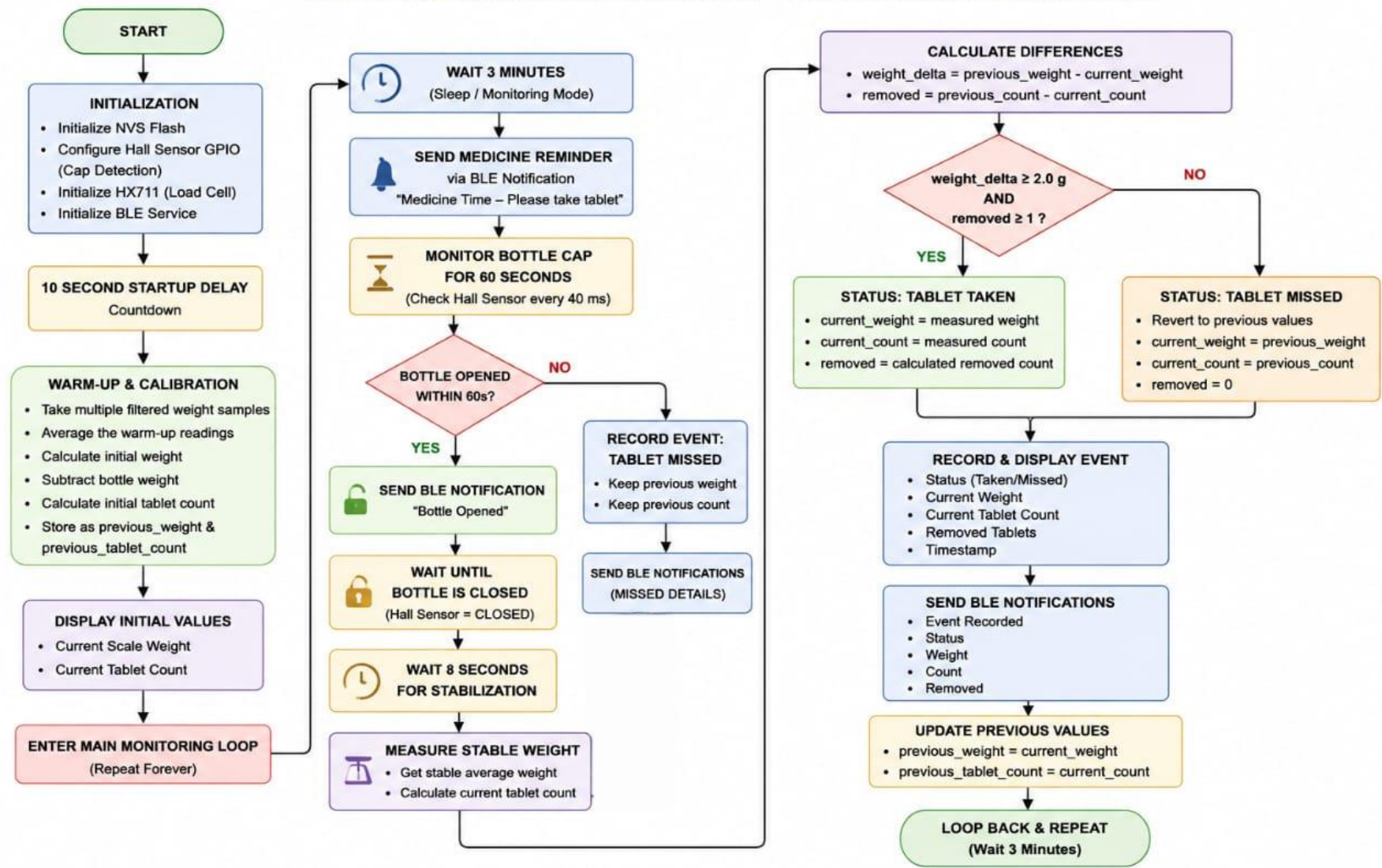
● Hardware Abstraction Layer (HAL)

- Acts as an intermediate layer between software drivers and hardware components.
- Provides standardized interfaces for **Timer, GPIO, HX711, NVS Flash, and BLE** operations.
- Simplifies communication between the application layer and hardware devices.
- Hides hardware-specific implementation details, improving portability and maintainability.
- Enables easy hardware upgrades or modifications without affecting the application logic.
- Ensures consistent and reliable access to hardware resources across the system.

● Hardware Layer

- **ESP32 Microcontroller** – The main processing unit responsible for data acquisition, decision-making, storage, and BLE communication.
- **Hall Effect Sensor** – Detects bottle cap opening and closing events using magnetic field changes.
- **HX711 Load Cell Amplifier** – Amplifies and converts the load cell's analog signals into digital data for the ESP32.
- **Load Cell** – Measures the bottle's weight to determine tablet removal and estimate medication consumption.
- **Pill Bottle Mechanism** – The physical container that stores the medication and integrates all sensing components.
- **BLE Module (ESP32 Built-in)** – Enables wireless communication with smartphones for notifications and status updates.
- **Power Supply/Battery** – Provides power to the ESP32 and sensing modules for continuous operation.

MEDICINE MONITORING SYSTEM – ALGORITHM FLOWCHART



WIRELESS COMMUNICATION (BLE) – SMART PILL BOTTLE



Demo Video

```
I (227) heap_init: AT 600000 len 00001F08 (7 Kib): MTCRAM
I (283) spi_flash: detected chip: gd
I (285) spi_flash: flash id: d10
W (288) spi_flash: Detected size(0192K) larger than the size in the binary image header(2048K). Using the size in the binary image header.
I (401) sleep_gpio: configure to isolate all GPIO pins in sleep state
I (407) sleep_gpio: enable automatic switching of GPIO sleep configuration
I (422) main_task: Started on CPU0
I (432) main_task: calling app.main()
I (342) BLE_INIT: BT controller compile version [1bb2f50]
I (342) BLE_INIT: Using main XTAL as clock source
I (442) BLE_INIT: Feature Config: ADV1, BLE_S01, DPM1, SCAN1, CEA10, SMP11, CONNECT11
I (452) BLE_INIT: Bluetooth MAC: 1C:DB:D4:44:58:16
I (452) phy_init: phy_version 711,97bcf0a2,Aug 25 2025,10:04:10
I (412) HISMILE: GAP procedure initiated: stop advertising.
I (412) HISMILE: GAP procedure initiated: advertise.
I (412) HISMILE: disc_mode=2
I (412) HISMILE: adv_channel_map=0 adv_tx_pwr=0 adv_filter_policy=0 adv_interval_min=0 adv_interval_max=0
I (422) HISMILE:
I (422) BLE: Advertising Started
I (422) BLE: BLE Initialized
Starting scale... 10 seconds
Starting scale... 8 seconds
Starting scale... 8 seconds
Starting scale... 7 seconds
Starting scale... 6 seconds
Starting scale... 5 seconds
Starting scale... 4 seconds
Starting scale... 3 seconds
Starting scale... 2 seconds
Starting scale... 1 seconds

Current Scale Weight : 36.29 g
Current Tablet Count : 5

=====
System entering sleep track mode. Monitoring window opens in 3 minutes...
```

DevicesDISCONNECT

EDADVERTISERSMART P...BOTTLE1C:DB:D4:44:58:16

CONNECT...BOONEDCLIENTSERVER

Generic Access
UUID: 0x1800
PRIMARY SERVICE

Generic Attribute
UUID: 0x1801
PRIMARY SERVICE

Unknown Service
UUID: 12345678-1234-5678-1234-567812345678
PRIMARY SERVICE

Unknown Characteristic
UUID: 87654321-4321-8765-4321-8765...
Properties: NOTIFY, READ
Descriptors:
Client Characteristic Configuration
UUID: 0x2902
Value: Notifications enabled

Thank you